

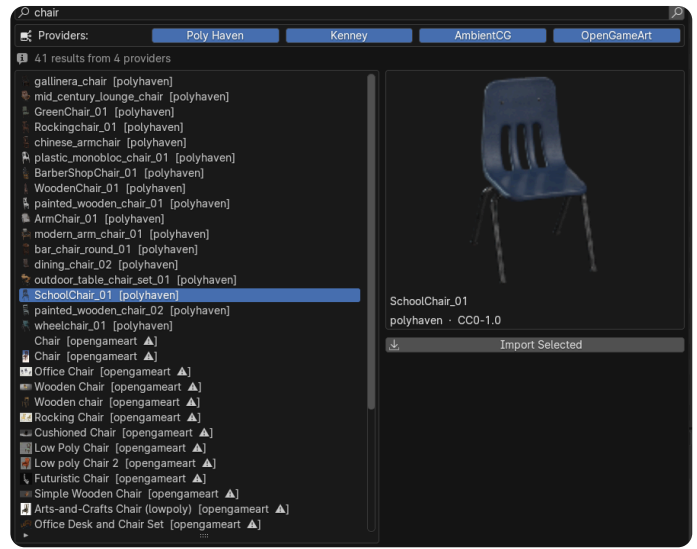
WF Asset Browser — User Manual

The WF Asset Browser is a Blender sidebar panel that searches licensed 3D assets across multiple online providers and records the **provenance** of everything you import — where it came from, who made it, under what terms, and what attribution you owe. Whether you ultimately accept or reject a given licence is a policy choice you configure per project; the tool's job is to make that choice informed and auditable.

Quick start

1. Open any `.blend` file, or run from a project directory that contains (or has an ancestor with) a `licence_policy.toml`.
2. Press **N** in the 3D Viewport to open the sidebar and click the **WF** tab. (Also available via **File** → **Import** → **WF Asset Browser**.)
3. Type a query (`barrel` , `tree` , `crate`) and press the search icon or **Enter**.
4. Pick a result and click **Import Selected**.

That's it for CC0 assets (the default). Anything more — enabling Sketchfab, accepting CC-BY, managing waivers — is covered in the sections below.



The policy file

The policy file is yours. The examples throughout this manual show the WF project's configuration for a commercial game. An open-source project might accept CC-BY-SA everywhere; a personal or hobbyist project might accept everything. The fallback (CC0 only) is conservative by design — configure it for your project.

All filtering is driven by a `licence_policy.toml` file. The plugin walks up from the active `.blend` file's directory until it finds this file; the first one found wins. The path shown in the panel under **Policy:** tells you which file is in effect.

If no file is found the panel shows:

```
Policy: (fallback – no licence_policy.toml found)
Accept: CC0-1.0
```

The fallback is intentional: new checkouts work immediately, but only the safest possible licence class passes. **Before you import anything other than CC0 assets, make sure the panel shows your project**

policy file, not the fallback.

Policy file anatomy

```
require_attribution_credits = true    # attribution strings must appear somewhere
```

```
[[licence]]
id      = "CC0-1.0"
status  = "accept"                  # this licence passes the filter
reason  = "public domain"
```

```
[[licence]]
id      = "CC-BY-4.0"
status  = "reject-default"         # blocked unless a per-asset [[waiver]] exists
reason  = "attribution required — waiver available"
```

```
[[licence]]
id      = "CC-BY-SA-4.0"
status  = "reject"                 # hard block, no waiver possible
reason  = "share-alike incompatible with commercial distribution"
```

status values:

Value	Meaning
accept	Assets with this licence appear in results and can be imported
reject-default	Blocked unless a [[waiver]] record exists for the specific asset
reject	Never passes — no waiver possible

Provenance — what the tool records

Every imported asset gets a `manifest.json` placed alongside it, even CC0 assets where attribution is not legally required. The record of origin is always there.

```
{
  "licence_id":      "CC0-1.0",
  "attribution_required": false,
  "attribution_string": "",
  "licence_url":     "https://creativecommons.org/publicdomain/zero/1.0/",
  "provider":        "polyhaven",
  "provider_id":     "oak-tree-01",
  "download_date":   "2026-04-29",
  "original_url":    "https://polyhaven.com/a/oak-tree-01",
  "download_url":    "https://cdn.polyhaven.com/...",
  "derived_from":    []
}
```

For CC-BY assets: `attribution_required` is `true` and `attribution_string` contains the credit line the author requests (e.g. "Oak Tree" by Jane Smith (sketchfab.com/...)). This string is what your credits screen must display — no guessing, no manual lookup.

The `derived_from` field tracks remix chains — if asset B is a modification of asset A, that relationship is preserved in the manifest.

Never delete `manifest.json` . It is the legal record of where the asset came from and under what terms. The future `wf_audit` CI tool will validate every asset in the level against its manifest automatically.

Attribution obligations audit

To see every asset in the project that requires attribution:

```
grep -rl '"attribution_required": true' wflevels/*/assets/
```

To extract just the credit strings for your credits screen:

```
grep -h '"attribution_string"' wflevels/*/assets/*/manifest.json | \
  grep -v '""' | sort -u
```

Run this before any release. Every `attribution_string` in the output is a line your credits screen must display.

Providers and their licence tiers

Each provider is a toggle in the **Providers:** box. The default-on set (grey background) are all CC0-only sources; Sketchfab (off by default) introduces mixed licences.

Provider	Licences available	Notes
Poly Haven	CC0-1.0	All assets; no API key
Kenney	CC0-1.0	Curated static catalog; no API key
AmbientCG	CC0-1.0	Textures and materials; no API key
Quaternius	CC0-1.0	Curated static catalog; no API key
OpenGameArt	CC0-1.0 subset	Open-upload bazaar — see ⚠ note below
Sketchfab	CC0 · CC-BY · CC-BY-SA · CC-BY-NC · CC-BY-ND · Editorial · Paid RF	Requires API key; see §Sketchfab

⚠ OpenGameArt trust level

OpenGameArt is an open-upload site. Assets are pre-filtered to CC0 by the provider API and re-filtered by the policy engine, but the accuracy of user-submitted licence claims is weaker than on curated sources. Results from OpenGameArt show a ⚠ warning glyph next to the licence label. Review them with extra care before shipping.

Licence tiers — what they mean and what they cost you

Licence	Commercial	Attribution	Derivatives	WF project default
CC0-1.0	✓	not required	✓	✓ accept
CC-BY-4.0	✓	required	✓	~ waiver only (credits screen required)
CC-BY-SA-4.0	✓	required	share-alike	✗ blocked — share-alike incompatible
CC-BY-NC-4.0	✗	required	✓	✗ blocked — no commercial use
CC-BY-NC-SA-4.0	✗	required	share-alike	✗ blocked — no commercial use
CC-BY-ND-4.0	✓	required	✗	✗ blocked — no derivatives
CC-BY-NC-ND-4.0	✗	required	✗	✗ blocked — no commercial use, no derivatives
royalty-free	✓	not required	✓	opt-in (add to policy; Sketchfab paid only)
editorial-only	✗	—	—	✗ blocked — press use only
unknown	?	?	?	✗ blocked — always fail-closed

CC0-1.0 — Public domain · no restrictions · no attribution

The green  icon in the results list. The author waives all rights; you can use, modify, and ship these assets with zero obligations.

What the tool records: Even though attribution is not required, the plugin writes the author's name and source URL to `manifest.json` for traceability.

WF project default: `accept` .

All five default providers serve CC0-only assets. If you leave the default provider set and the default policy, every result you see is CC0.

CC-BY-4.0 — Attribution required

The yellow  icon. The author allows any use — commercial, derivative works, redistribution — as long as you credit them.

What the tool records: `attribution_string` contains the exact credit line the author requests. `attribution_required` is set to `true` . This is what your credits screen must display.

WF project default: `reject-default` . CC-BY assets do *not* appear in results unless you either:

- Set the licence to `accept` in `licence_policy.toml` (accepts CC-BY globally); or

- Add a `[[waiver]]` for the specific asset (accepts one asset despite the default rejection).

When you enable CC-BY globally or via waiver, you are committing the project to maintaining a credits screen that displays the `attribution_string` for every imported CC-BY asset. Run the attribution audit command (see §Provenance) before any release.

When it suits other projects: Fine for any project that can maintain a credits screen — which is most projects. Set `status = "accept"` to enable globally.

How to add a global accept:

```
[[licence]]
id      = "CC-BY-4.0"
status = "accept"
reason = "credits screen maintained"
```

How to add a per-asset waiver (the safer choice for one-off assets):

```
[[waiver]]
asset_id    = "sketchfab/abc123uid"
licence_id  = "CC-BY-4.0"
approved_by = "wbnorris"
approved_at = "2026-04-29"
reason      = "stylistic fit; credits screen updated with attribution_string"
```

Or via the CLI:

```
wf-asset policy add-waiver sketchfab/abc123uid CC-BY-4.0 \
--reason "stylistic fit; credits updated"
```

CC-BY-SA-4.0 — Attribution + ShareAlike

What it requires: Derivatives must be released under the same CC-BY-SA terms. The tool records `attribution_string` and `licence_url` for any asset you accept.

WF project default: hard block. No waiver path. ShareAlike requires any derivative work to be released under the same licence. For a commercial game, that means releasing the entire project under CC-BY-SA — incompatible with commercial distribution.

When it suits other projects: Perfect for open-source games that want remixed assets to stay open. If your project is open-source, set this to `accept`.

Assets with this licence are silently stripped from results under the WF default policy. If you explicitly need one and your project is commercial, that decision requires legal review and is outside the scope of the policy file.

CC-BY-NC-4.0 and CC-BY-NC-SA-4.0 — NonCommercial

What they require: No commercial use. The tool records full provenance regardless of policy outcome.

WF project default: hard block. No waiver path. NonCommercial clauses prohibit any revenue-generating use, including game sales on any storefront.

When they suit other projects: Fine for personal, hobby, or academic work with no monetisation. Set to `accept` in your policy file.

CC-BY-ND-4.0 and CC-BY-NC-ND-4.0 — NoDerivatives

What they require: No derivative works. The tool records provenance and flags `attribution_required = true`.

WF project default: hard block. ND licences prohibit creating derivative works — which includes format conversion, scaling, re-texturing, and any modification made during a normal asset pipeline. Almost never appropriate for game assets.

When they suit other projects: Rare — background decoration used exactly as-is. A narrow waiver path via `[[waiver]]` is possible if needed.

royalty-free — Sketchfab Standard (paid)

The ► icon. Assets purchased through Sketchfab's store under the Standard licence: one-time purchase, unlimited commercial use, no attribution required. Comparable to a stock photo royalty-free licence.

Default policy status: not listed — blocked by default. To accept purchased Sketchfab Standard assets, add to `licence_policy.toml`:

```
[[licence]]
id      = "royalty-free"
status  = "accept"
reason  = "purchased Sketchfab Standard; one-time fee logged in manifest"
```

The plugin will write the purchase price as \$0 in `manifest.json` (it has no access to your payment receipt). Keep your own record of the purchase date and order number.

editorial-only — Press/editorial use

Hard block. These licences permit press coverage and editorial illustration but explicitly exclude entertainment products and games. Cannot be waived.

unknown — Unrecognised licence string

Always blocked. The provider returned a licence string the engine could not map to any known `LicenceId`. Check the provider's site for the actual licence, then either add a mapping (requires a code change in `src/licence.rs`) or add a waiver referencing the correct licence ID.

Reading the results list

The screenshot at the top of this document shows a live search for "chair" across four providers. Key things to notice:

- **Poly Haven and Kenney results** appear at the top of the list — curated, all CC0, no warning markers.

- **OpenGameArt results** (lower portion) carry a  after the licence label — the lower-trust flag for open-upload sources.
- The **selected row** (`SchoolChair_01` [polyhaven]) shows a thumbnail and `polyhaven · CC0-1.0` in the detail pane on the right.
- The **Import Selected** button is active because the selected asset passes the current policy.

Each row in the results list shows:

[thumb] Asset Title
Provider · LICENCE-ID [icon]

Icon	Meaning
 (green)	CC0-1.0 — public domain, zero obligations
 (yellow)	CC-BY variant — attribution required
	royalty-free (paid Sketchfab Standard)
 (red)	editorial-only or unknown — policy will block import
 (after licence)	OpenGameArt lower-trust source

The **Filter** dropdown above the list narrows what's shown:

Filter	Shows
All	Everything that passes the policy
CC0 only	CC0-1.0 exclusively
CC (free)	Any CC licence — CC0, CC-BY, CC-BY-SA, etc.
Paid RF	royalty-free purchased assets only

Note: the filter is applied *after* the policy. Setting Filter to "CC (free)" while CC-BY-SA is `reject` in the policy will show CC0 and any CC-BY you've explicitly accepted, but *not* CC-BY-SA.

Sketchfab setup

Sketchfab is the only provider that requires authentication. Search works without a key (you can browse and preview); download requires a Bearer API token.

How to add your key:

1. Visit `sketchfab.com/settings#api-token` and copy your token.
2. In Blender: **Edit** → **Preferences** → **Add-ons**, search for **World Foundry**.
3. Paste the token into the **Sketchfab API Key** field.

The field is stored in Blender's user preferences (not in any project file or environment variable). It is masked as a password field in the UI.

Enable Sketchfab in the panel:

In the **Providers:** box, toggle **Sketchfab** on. If no API key has been set, the panel shows an information note:

```
No API key – search works, download won't.  
Set key in Edit > Preferences > Add-ons > World Foundry.
```

With a key set, enabling the toggle and searching returns Sketchfab results alongside the CC0 providers. Results carry per-asset licence badges — a single Sketchfab search may return CC0, CC-BY, Editorial, and paid assets all at once; the policy filter strips everything that isn't accepted.

For the CLI (`wf-asset` binary), set the environment variable instead:

```
export WF_SKETCHFAB_API_KEY="your-token-here"  
wf-asset search "medieval chair" --provider sketchfab
```

Importing an asset

1. Run a search and select a result row.
2. Click **Import Selected**.

The plugin:

- Downloads the asset to `assets/<provider>/<asset-id>/` relative to the active `.blend` file (created automatically).
- Writes `manifest.json` alongside the asset file (see §Provenance).
- Invokes Blender's built-in glTF importer.
- Places the imported mesh at the **3D cursor**.
- Sets a `wf_schema_path` custom property on the imported objects pointing to `wfsource/source/oas/statplat.oas` (the default static-platform schema).

The status bar at the top of the panel shows progress and the final result:

```
Imported "Oak Tree" (CC0-1.0) from Poly Haven; manifest written alongside asset
```

CLI reference

The `wf-asset` binary (built from `wftools/wf_asset_provider/`) lets you use the asset system from the terminal without opening Blender.

```
# Search (reads policy from current directory upward)  
wf-asset search "tree" --provider polyhaven --limit 10
```



```
# Download directly
wf-asset download polyhaven/oak-tree-01 --dest wflevels/test/assets

# Show the active policy
wf-asset policy show

# List registered providers
wf-asset providers list
```

Set `WF_SKETCHFAB_API_KEY` in the environment to enable Sketchfab downloads.

Troubleshooting

"wf_core not found"

The compiled native library (`wf_core.so`) is missing from the add-on directory. Build and install it:

```
task blender-install
```

This runs `maturin build --release` if the wheel is missing, then copies `wf_core.so` into the add-on directory and symlinks all Python files.

Results I expected are missing

Check the **Policy:** line in the panel. If it says (fallback) , the plugin couldn't find a `licence_policy.toml` and is accepting only CC0. Open the `.blend` file from a directory that has `licence_policy.toml` in its ancestor tree.

Check the **Filter** dropdown — if it is set to "CC0 only" and you are searching Sketchfab, non-CC0 results will be hidden even if they passed the policy.

"authentication required for sketchfab"

The Sketchfab toggle is on and you attempted to download, but no API key is configured. See §Sketchfab setup.

A Sketchfab asset I bought is blocked

The `royalty-free` licence ID is not in your `licence_policy.toml` . Add:

```
[[licence]]
id      = "royalty-free"
status  = "accept"
reason  = "purchased Sketchfab Standard assets"
```

An asset shows × (red icon) in the list

Its licence is `editorial-only` or `unknown` . Editorial-only is a hard block with no waiver path. For `unknown` , check the provider's page for the actual licence text; if it maps to a known ID, file an issue so the `LicenceId::from_raw()` table can be updated.

Licence decision flowchart

Is the licence CC0-1.0?

└ Yes → Import freely. No obligations.

└ No → Is it CC-BY-4.0?

└ Yes → Is it in policy as accept or waived?

└ Yes → Import. Add attribution_string to credits screen.

└ No → Add waiver in licence_policy.toml, then import.

└ No → Is it CC-BY-SA, CC-BY-NC, CC-BY-ND, GPL, or editorial?

└ Yes → With the WF project default policy: hard block.

For open-source projects: configure your policy.

For hobby/non-commercial: configure your policy.

└ No → Is it royalty-free (Sketchfab paid Standard)?

└ Yes → Add "royalty-free" as accept in policy. Import.

Keep your purchase receipt separately.

└ No → licence is unknown.

→ Check the provider's page. If recognisable, file an issue to extend `LicenceId::from_raw()`. Otherwise skip.

Design reference

- [Asset Browser Plugin Plan](#) — v1 implementation plan: architecture, provider scope, import flow, effort estimate.
- [Sketchfab + Commercial Providers Plan](#) — v2 implementation plan: credentials layer, Sketchfab API, licence-filter UI.
- [Level Construction Tooling Investigation](#) — Seam 5 context: why licence is a gate, not metadata; full provider shortlist; phased roadmap.